

Unit-2 - Analytical Questions

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1) Plaintext : 0123456789 ABCDEF

Key : 133457799BBCDFF1

DES Encryption Process :

Step:	Description
Initial Permutation	The 64-bit plaintext is rearranged using DES IP table
Key scheduling	The 64-bit key is reduced to 56 bits key reasoning parity bits.
Round 1 Feistel operation	The plaintext is divided into left (L_0) and right (R_0) value. R_0 is expanded to 48 bits, XORed with K_1 , passed through S-lines and P-lines, the XORed with L_0 to produce new right half.
Round 2-16	The same feistel opr. is repeated using round keys K_2 to K_{16}

2. Plaintext : 1111222233334444

Key : AABBO9182736CCDD

Step	Description
IP	The plaintext undergoes DES IP

Key Scheduling	16, 48 bit round keys are generated from the given by using PC-1, left shifts, and PC-2
Ferstel Rounds	The data block passes through 16 Ferstel rounds consisting of expansion, XOR, S-box rule.
Final P	The inverse permutation is applied to obtain ciphertext
Result	The ciphertext is obtained after final DES permutation.

3. Plaintext : 123456ABCD132536
 Key : AABBBCCDDEEFF0011
- * Perform 16 ferstel rounds
 - * Swap the left and right halves
 - * Apply FP
 - * Obtain encryption 64 bit ciphertext.
 - * Apply IP
 - * generate 16 rounds keys from given key
4. Plaintext : FEDCBA9876543210
 Key : 0F1571C947D9E859

Step	Description
IP	Rearrange plaintext accordingly to the IP table
Key schedule	generate 16 round keys using PC-1, circular left shifts, & PC-2
Round operations	Each round performs expansion $\rightarrow$ XOR $\rightarrow$ S-box rule. $\rightarrow$ P-box $P \rightarrow$ XOR with left half

Plaintext : 02468 ACEECA86420

Key : 0E329232EAD0D73

DES Encryption:

1. Apply Initial Permutation
2. gen 16 subkeys
3. Perform 16 feistel rounds
4. Swap halves
5. Apply FP
6. Obtain the encryption ciphertext.

6. Plaintext : 0123456789ABCDEF

Key : FEDCBA98765432100123456789ABCDEF

Step	Description
Init	Init the P-array and four S-boxes using blowfish
Key expansion	XOR the user key with P-array & repeatedly encrypt
Final swap	Swap the left and right halves after the 16th round
Final XOR	XOR with the last two P-array entries to produce the ciphertext.

7. Plaintext : 123456789ABCDEF

Key : 00112233445566778899AABBCCDDEEFF

Process:

- * Init P-array and S-boxes
- * expand 128-bit key
- * Perform 16 feistel rounds

8. Plaintext : FEDCBA9876543210

Key : AABBBCCODEEFF00112233445566778899

Steps:

1. Init P-arrays and S-boxes
2. Perform expansion
3. Divide plaintext into two halves

4. execute sixteen feistel rounds

5. Apply for final swap

6. obtain ciphertext

9. PT = 89ABCDEF01234567

Key : 1234567890ABCDEF FEDCBA0987654321

1. P-array/S-box setup from key
2. feistel rounds 1-16 (F-function = s-box additions/xors)
3. Final P_L/P_R XOR

Ciphertext = 7FF938AF44E2B9C7

10. PT = ABCDEF1234567890, P

Key = 0F1E2D3C4B5A69788796A5B4C3D2E1F0

1. Initialize P-array/S-boxes with key
2. Run all 16 feistel rounds on plaintext halves
3. Final whitening (P_L, P_R) gives ciphertext = D89658661A57DB9.